

MSc Strategic Marketing

MARKETING ANALYTICS

(BUSI70278)

XXday XX December 20XX; XX:00- XX:00

CLOSED BOOK

Instructions

Please read the questions carefully and answer **ALL questions** provided, including **ALL sub-questions**.

The use of calculators is allowed.

Write down the intermediary steps of your calculations. Do not just notate final numbers.

If you need to make assumptions, explain these clearly.

QUESTION 1

[Total: 65 marks]

A brand manager of brand ABC has access to 150 weeks of the following data, and wants to get an understanding of its sales drivers:

Variable	Definition	Average
Online advertising _t	Total spending in pounds on online advertising in week t	83,520.30
Offline advertising _t	Total spending in pounds on offline advertising in week t	22,730.45
Regular price _t	Price per unit in pounds in week t	125.35
Sales _t	Total sales in units in week t	175,368.17

You estimate the following log-log regression model, after mean-centering all three predictor variables:

$$\log(\text{Sales}_t) = \beta_0 + \beta_1 \log(\text{Online advertising}_t) + \beta_2 \log(\text{Offline advertising}_t) + \beta_3 \log(\text{Regular price}_t) + \varepsilon_t$$

The results are:

Variable	Estimate	P-value
Intercept (β_0)	12.12	<.01
Online advertising _t (β_1)	0.23	0.01
Offline advertising _t (β_2)	0.12	0.02
Regular price _t (β_3)	-1.35	<.01

- a) What is the expected percentage change in sales units if you increase online advertising by 10%?

(8 marks)

- b) What is the expected percentage change in sales units if you decrease the regular price by 10%?

(8 marks)

- c) Imagine you have a 600,000-pound advertising budget available. Based on the ratio of elasticities method, how would you allocate your advertising budget between online versus offline advertising? Should the brand allocate a larger or smaller share of its advertising budget to online advertising than it is currently doing?

(10 marks)

- d) You learn from the brand manager that the brand's offline advertising mostly consists of brand building campaigns to deliver long-term value for the brand, whereas the brand's online advertising mostly consists of search and display advertising to drive short-term sales. Using the available data, discuss how could use this insight to improve the regression model.

(12 marks)

- e) You wonder whether online advertising enhances the effectiveness of offline advertising. Using the available data, discuss how would you answer this question.

(12 marks)

- f) The brand manager informs you that one of the brand's key competitors reduced its prices during the 150 weeks in the sample, which the brand manager believes to have reduced brand ABC's sales. These competitive price reductions typically occurred during the same weeks when brand ABC reduced its prices. Discuss whether and how not including the price of this competitor in the model may or may not bias the observed estimate of regular price.

(15 marks)

QUESTION 2

[Total: 35 marks]

An online retailer is looking incentivize purchases by sending discounted coupons to registered users. The brand would like to explore offering a discounted coupon to a selected group of consumers.

The retailer selects 2,000 users that visited the website in the last month without making a transaction. A random selection of 50% of users are offered a non-transferable 10% discount valid for 1 month on their next purchase.

The retailer collects data on total spending in pounds of all users during the 1-month test period, and estimates a simple *linear* regression model (*not a log-log model, and without mean-centering*), using total spending in pounds of a user as the response variable, and a binary indicator ("Coupon") that equals 1 if a user was offered a discount and 0 otherwise as the only predictor variable.

The results are:

Variable	Estimate	P-value
Intercept	7.55	<.01
Coupon	0.45	0.47

- a) Interpret the two estimates, and discuss spending levels among the two groups of customers as well as how the coupon has affected spending.

(15 marks)

- b) Next, you explore the role of customer heterogeneity in the effect of the coupon. You interact the coupon effect ('Coupon') with shopping cart status ('Shopping_cart'). 'Shopping_cart' equals 1 if the user had something in their shopping cart at the start of the test period, and 0 otherwise.

The results are:

Variable	Estimate	P-value
Intercept	7.55	<.01
Coupon	2.45	<.01
Shopping_cart	1.75	0.02
Coupon × Shopping_cart	-2.45	0.01

Interpret and discuss the role of shopping cart status in the effect of the coupon.

(10 marks)

- c) Imagine the above test was done based on data collected from a period during which inflation was stable at 2%. Imagine that inflation recently sharply increased to 15%. Should the retailer still trust the results generated by this test? Why (not)? What would you advise the retailer to do?

(10 marks)

End of Paper